



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta del 21/11/2023*

PROBLEM - We observe N samples of the following discrete-time real-valued w.s.s. random process:

$$X[n] = S[n] + W[n] = A + W[n], \quad n = 0, 1, 2, \dots, N-1,$$

where A is a constant signal modelled as a Gaussian distributed random variable:

$$f_A(a) = \frac{1}{\sqrt{2\pi\sigma_A^2}} e^{-\frac{(a-\eta_A)^2}{2\sigma_A^2}}, \quad \text{where } E\{A\} = \eta_A, \quad E\{(A-\eta_A)^2\} = \sigma_A^2,$$

and $W[n]$ is additive white Gaussian noise (AWGN), zero mean, with power σ_w^2 . Define γ as the ratio of the signal and noise variances, i.e. $\gamma = \text{var}\{S[n]\} / \text{var}\{W[n]\}$.

Derive:

(1) the mean vector $\mathbf{m}_X$ and the covariance matrix $\mathbf{C}_X$ of the observed data vector:

$$\mathbf{X} = [X[0] \ X[1] \ \dots \ X[N-1]]^T;$$

(2) the autocorrelation function (ACF) and the Power Spectral Density (PSD) of the observed process $X[n]$;

(3) the Maximum A Posteriori (MAP) estimator of the random parameter A given the observation of $\mathbf{X}$ and express it as a function of γ , of the sample size N , the noise power σ_w^2 , and of the Maximum Likelihood (ML) estimator of A (which in this case is the well-known Sample Mean Estimator); discuss what happens to $\hat{A}_{MAP}$ when $N \rightarrow \infty$ or $\gamma \rightarrow \infty$ (keeping constant σ_w^2);

(4) the bias and the mean square error (MSE) of the MAP estimator of A ; discuss what happens to $MSE\{\hat{A}_{MAP}\}$ when $N \rightarrow \infty$ or $\gamma \rightarrow \infty$ (keeping constant σ_w^2); plot $MSE\{\hat{A}_{MAP}\}$ as a function of γ and as a function of the sample size N ;

(5) the Bayesian Cramér-Rao bound (BCRB) on the estimate of A and determine whether the MAP estimator is efficient or not.



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Solution

(1) Mean vector and covariance matrix of the data vector $\mathbf{X}$

$$\mathbf{m}_X = E\{\mathbf{X}\} = E\{A\mathbf{1} + \mathbf{W}\} = E\{A\}\mathbf{1} = \eta_A \mathbf{1}$$

$$\begin{aligned} \mathbf{C}_X &= E\{(\mathbf{X} - \mathbf{m}_X)(\mathbf{X} - \mathbf{m}_X)^T\} = E\{(\mathbf{X} - \eta_A \mathbf{1})(\mathbf{X} - \eta_A \mathbf{1})^T\} \\ &= E\{((A - \eta_A)\mathbf{1} + \mathbf{W})((A - \eta_A)\mathbf{1} + \mathbf{W})^T\} = \text{var}\{A\}\mathbf{1}\mathbf{1}^T + \mathbf{C}_W = \sigma_A^2 \mathbf{1}\mathbf{1}^T + \sigma_W^2 \mathbf{I} \end{aligned}$$

(2) ACF and PSD of the process $X[n]$

$$\gamma = \text{var}\{S[n]\} / \text{var}\{W[n]\} = \text{var}\{A\} / \sigma_W^2 = \sigma_A^2 / \sigma_W^2$$

$$R_X[m] = E\{X[n]X[n+m]\} = E\{(A + W[n])(A + W[n-m])\} = E\{A^2\} + R_W[m] = \eta_A^2 + \sigma_A^2 + \sigma_W^2 \delta[m]$$

$$S_X(e^{j2\pi F}) = FT\{R_X[m]\} = FT\{\eta_A^2 + \sigma_A^2\} + FT\{\sigma_W^2 \delta[m]\} = 2\pi(\eta_A^2 + \sigma_A^2)\delta(e^{j2\pi F} - 1) + \sigma_W^2$$

(3) Maximum A Posteriori (MAP) estimator of A

$$\hat{A}_{MAP} = \arg \max_a f_{A|X}(a|\mathbf{x}) = \arg \max_a f_X(\mathbf{x}, a) = \arg \max_a \ln f_X(\mathbf{x}, a)$$

$$f_X(\mathbf{x}, a) = f_X(\mathbf{x}|a)f_A(a) = f_X(\mathbf{x}|a) \cdot \frac{1}{\sqrt{2\pi\sigma_A^2}} e^{-\frac{(a-\eta_A)^2}{2\sigma_A^2}}$$

Taking into account that the noise samples are IID, the conditional pdf (in scalar and vector notation) is given by:

$$\begin{aligned} f_X(\mathbf{x}|a) &= \prod_{n=0}^{N-1} \frac{1}{\sqrt{2\pi\sigma_W^2}} \exp\left(-\frac{(X[n]-a)^2}{2\sigma_W^2}\right) = \frac{1}{\sqrt{(2\pi)^N |\sigma_W^2 \mathbf{I}|}} \exp\left(-\frac{1}{2}(\mathbf{x} - a\mathbf{1})^T (\sigma_W^2 \mathbf{I})^{-1} (\mathbf{x} - a\mathbf{1})\right) \\ &= \frac{1}{(\sqrt{2\pi\sigma_W^2})^N} \exp\left(-\frac{1}{2\sigma_W^2}(\mathbf{x} - a\mathbf{1})^T (\mathbf{x} - a\mathbf{1})\right) \end{aligned}$$

Hence, the logarithm of the joint pdf of $\mathbf{X}$ and A can be expressed as:



$$\ln f_{\mathbf{x}}(\mathbf{x}, a) = -\frac{N}{2} \ln(2\pi\sigma_w^2) - \frac{1}{2\sigma_w^2} (\mathbf{x} - a\mathbf{1})^T (\mathbf{x} - a\mathbf{1}) - \frac{1}{2} \ln(2\pi\sigma_A^2) - \frac{(a - \eta_A)^2}{2\sigma_A^2}$$

that can be rearranged as:

$$\begin{aligned} \ln f_{\mathbf{x}}(\mathbf{x}, a) &= k - \frac{1}{2\sigma_w^2} (\mathbf{x}^T \mathbf{x} - a\mathbf{1}^T \mathbf{x} - a\mathbf{x}^T \mathbf{1} + a^2 \mathbf{1}^T \mathbf{1}) - \frac{(a - \eta_A)^2}{2\sigma_A^2} = k - \left(\frac{N}{2\sigma_w^2} + \frac{1}{2\sigma_A^2} \right) a^2 + \left(\frac{1}{\sigma_w^2} \mathbf{1}^T \mathbf{x} + \frac{\eta_A}{\sigma_A^2} \right) a \\ &= k - \frac{N}{2\sigma_w^2} \left(1 + \frac{\sigma_w^2}{N\sigma_A^2} \right) a^2 + \frac{N}{\sigma_w^2} \left(\frac{1}{N} \mathbf{1}^T \mathbf{x} + \frac{\sigma_w^2}{N\sigma_A^2} \eta_A \right) a \\ &= k - \frac{N}{2\sigma_w^2} \left(1 + \frac{1}{N\gamma} \right) a^2 + \frac{N}{\sigma_w^2} \left(\frac{1}{N} \mathbf{1}^T \mathbf{x} + \frac{1}{N\gamma} \eta_A \right) a \end{aligned}$$

where k includes all terms that do not depend on a , and so are irrelevant.

The MAP estimator can be obtained by solving the MAP equation:

$$\frac{d \ln f_{\mathbf{x}}(\mathbf{x}, a)}{da} = -\frac{N}{\sigma_w^2} \left(1 + \frac{1}{N\gamma} \right) a + \frac{N}{\sigma_w^2} \left(\frac{1}{N} \mathbf{1}^T \mathbf{x} + \frac{1}{N\gamma} \eta_A \right) = 0$$

The solution of this 1st order linear equation is:

$$\begin{aligned} a &= \frac{1}{\left(1 + \frac{1}{N\gamma} \right)} \left(\frac{1}{N} \mathbf{1}^T \mathbf{x} + \frac{1}{N\gamma} \eta_A \right) = \frac{N\gamma}{1 + N\gamma} \left(\frac{1}{N} \mathbf{1}^T \mathbf{x} + \frac{1}{N\gamma} \eta_A \right) = \frac{N\gamma}{1 + N\gamma} \left(\frac{1}{N} \mathbf{1}^T \mathbf{x} \right) + \frac{1}{1 + N\gamma} \eta_A \\ &= \frac{N\gamma}{1 + N\gamma} \left(\frac{1}{N} \mathbf{1}^T \mathbf{x} \right) + \left(1 - \frac{N\gamma}{1 + N\gamma} \right) \eta_A = \alpha \hat{A}_{ML} + (1 - \alpha) \eta_A \end{aligned}$$

where we took into account that $\hat{A}_{ML} = \frac{1}{N} \mathbf{1}^T \mathbf{x} = \frac{1}{N} \sum_{n=0}^{N-1} X[n]$, and we defined $\alpha \triangleq \frac{N\gamma}{1 + N\gamma}$.

When $N \rightarrow \infty$ or $\gamma \rightarrow \infty$ (i.e. $\sigma_s^2 \rightarrow \infty$) we have: $\alpha \rightarrow 1$ so that $\hat{A}_{MAP} \rightarrow \hat{A}_{ML}$.

When $N \rightarrow 0$ or $\gamma \rightarrow 0$ (i.e. $\sigma_s^2 \rightarrow 0$) we have: $\alpha \rightarrow 0$ so that $\hat{A}_{MAP} \rightarrow \eta_A$, i.e. the a-priori mode (and mean) of A . Since $\sigma_s^2 = 0$, we have that $A = \eta_A$, so $\hat{A}_{MAP} = A$ and the estimation error is 0.



(4) Bias and MSE of the MAP estimator of A

$$\hat{A}_{MAP} = \alpha \hat{A}_{ML} + (1 - \alpha) \eta_A$$

$$\text{bias}\{\hat{A}_{MAP}\} = E\{A - \hat{A}_{MAP}\} = \eta_A - \alpha E\{\hat{A}_{ML}\} - (1 - \alpha) \eta_A = -\alpha E\{A\} + \alpha \eta_A = -\alpha (E\{A\} - \eta_A) = 0$$

Hence, the MAP estimator is **unbiased**.

$$\begin{aligned} MSE\{\hat{A}_{MAP}\} &= E\left\{\left(A - \hat{A}_{MAP}\right)^2\right\} = E\left\{\left(A - \alpha \hat{A}_{ML} - (1 - \alpha) \eta_A\right)^2\right\} \\ &= E\left\{\left(A - \alpha A + \alpha A - \alpha \hat{A}_{ML} - (1 - \alpha) \eta_A\right)^2\right\} = E\left\{\left[\alpha(A - \hat{A}_{ML}) + (1 - \alpha)(A - \eta_A)\right]^2\right\} \\ &= \alpha^2 E\left\{\left(A - \hat{A}_{ML}\right)^2\right\} + (1 - \alpha)^2 E\left\{\left(A - \eta_A\right)^2\right\} + 2\alpha(1 - \alpha) E\left\{\left(A - \hat{A}_{ML}\right)(A - \eta_A)\right\} \\ &= \alpha^2 \frac{\sigma_w^2}{N} + (1 - \alpha)^2 \sigma_A^2 = \frac{\sigma_w^2}{N} (\alpha^2 + (1 - \alpha)^2 N \gamma) = \frac{\sigma_w^2}{N} \left(\frac{(N \gamma)^2}{(N \gamma + 1)^2} + \frac{N \gamma}{(N \gamma + 1)^2} \right) \\ &= \frac{\sigma_w^2}{N} \left(\frac{N \gamma}{N \gamma + 1} \right) = \alpha \frac{\sigma_w^2}{N} \end{aligned}$$

We observe that $MSE\{\hat{A}_{MAP}\} = \alpha MSE\{\hat{A}_{ML}\}$, since it is well-known that for this estimation problem we have $MSE\{\hat{A}_{ML}\} = \frac{\sigma_w^2}{N}$.

Moreover, we can also express the MSE of the MMSE estimator as follows:

$$MSE\{\hat{A}_{MAP}\} = \frac{\sigma_w^2}{N} \left(\frac{N \gamma}{N \gamma + 1} \right) = \frac{\sigma_w^2 \gamma}{N \gamma + 1} = \frac{\sigma_s^2}{N \gamma + 1} = \left(1 - \frac{N \gamma}{N \gamma + 1} \right) \sigma_s^2 = (1 - \alpha) \sigma_s^2$$

When $N \rightarrow \infty$ or $\gamma \rightarrow \infty$ (i.e. $\sigma_s^2 \rightarrow \infty$) we have: $\alpha \rightarrow 1$, hence $MSE\{\hat{A}_{MAP}\} \rightarrow MSE\{\hat{A}_{ML}\}$.

When $N \rightarrow 0$ or $\gamma \rightarrow 0$ (i.e. $\sigma_s^2 \rightarrow 0$) we have: $\alpha \rightarrow 0$ so that $MSE\{\hat{A}_{MAP}\} \rightarrow \sigma_s^2$, which is the variance of a-priori pdf of A . Since $\sigma_s^2 = 0$, we have that in this case $MSE\{\hat{A}_{MAP}\} \rightarrow 0$, i.e. the estimation error is 0.



(5) Bayes Cramér-Rao bound (BCRB)

$$BCRB(A) = -\frac{1}{E\left\{\frac{d^2 \ln f_x(\mathbf{x}, a)}{da^2}\right\}}$$

$$\frac{d \ln f_x(\mathbf{x}, a)}{da} = -\frac{N}{\sigma_w^2} \left(1 + \frac{1}{N\gamma}\right) a + \frac{N}{\sigma_w^2} \left(\frac{1}{N} \mathbf{1}^T \mathbf{x} + \frac{1}{N\gamma} \eta_A\right), \quad \frac{d^2 \ln f_x(\mathbf{x}, a)}{da^2} = -\frac{N}{\sigma_w^2} \left(1 + \frac{1}{N\gamma}\right)$$

$$E\left\{\frac{d^2 \ln f_x(\mathbf{x}, a)}{da^2}\right\} = E\left\{-\frac{N}{\sigma_w^2} \left(1 + \frac{1}{N\gamma}\right)\right\} = -\frac{N}{\sigma_w^2} \left(1 + \frac{1}{N\gamma}\right)$$

$$BCRB(A) = -\frac{1}{-E\left\{\frac{d^2 \ln f_x(\mathbf{x}, a)}{da^2}\right\}} = \frac{\sigma_w^2}{N} \left(\frac{N\gamma}{N\gamma+1}\right) = \alpha \frac{\sigma_w^2}{N} = (1-\alpha) \sigma_s^2 = MSE\left\{\hat{A}_{MAP}\right\},$$

Hence, the MAP estimator is **efficient**.